Revision Games Pack — 1.3 Exchanging Data

OCR A Level Computer Science (H446) — Component 01, Section 1.3 (Compression, encryption & hashing; Databases; Networks; Web technologies)

Loop cards (dominoes)

How to play: Print and cut out the cards, then shuffle and deal one or more to each student. One person reads the "Who has…?" clue at the bottom of their card. Whoever holds the matching term reads it out, then reads *their* clue. Continue until you return to the start — the loop is closed and self-checking, so any mistake breaks the chain.

- **Card 1 — I have: RLE (Run Length Encoding).** Who has the lossless method that stores repeated patterns once and references them?
- **Card 2 — I have: Dictionary compression.** Who has the term for compression where data is permanently removed and cannot be recovered?
- **Card 3 — I have: Lossy compression.** Who has the one-way function that produces a fixed-length digest?
- **Card 4 — I have: Hashing.** Who has the encryption type that uses the same key to encrypt and decrypt?
- **Card 5 — I have: Symmetric encryption.** Who has the encryption type that uses a public and private key pair?
- **Card 6 — I have: Asymmetric encryption.** Who has the attribute(s) that uniquely identify each record in a table?
- **Card 7 — I have: Primary key.** Who has the primary key of another table used to link tables together?
- **Card 8 — I have: Foreign key.** Who has the process of organising data to reduce redundancy and anomalies?
- **Card 9 — I have: Normalisation.** Who has the rule requiring no partial dependencies (every non-key attribute depends on the whole key)?
- **Card 10 — I have: 2NF (Second Normal Form).** Who has the set of properties guaranteeing reliable database transactions (Atomicity, Consistency, Isolation, Durability)?
- **Card 11 — I have: ACID.** Who has the network where data is split into packets routed independently and reassembled at the destination?
- **Card 12 — I have: Packet switching.** Who has the system that translates domain names into IP addresses?
- **Card 13 — I have: DNS.** Who has the device that connects different networks together using IP addresses?
- **Card 14 — I have: Router.** Who has the device that connects devices within a single LAN using MAC addresses?
- **Card 15 — I have: Switch.** Who has the algorithm that ranks web pages by importance based on incoming links?
- **Card 16 — I have: PageRank.** Who has the lossless method that replaces runs of repeated values with a value and a count?

(Card 16 loops back to Card 1.)

Taboo cards

How to play: In pairs/teams, a player describes the **TERM** at the top of the card to their team **without saying any of the 4 banned words** (or obvious variations). The team has 60 seconds to guess as many as possible. One point per correct guess; opposing team watches for banned words.

Card 1 — HASHING Banned: one-way · password · digest · reversible

Card 2 — RLE Banned: run · repeated · count · lossless

Card 3 — FOREIGN KEY Banned: primary · link · table · reference

Card 4 — NORMALISATION Banned: redundancy · anomaly · table · 1NF

Card 5 — PACKET SWITCHING Banned: split · route · reassemble · internet

Card 6 — DNS Banned: domain · IP · address · translate

Card 7 — SYMMETRIC ENCRYPTION Banned: same · key · decrypt · distribution

Card 8 — PAGERANK Banned: link · importance · vote · ranking

Bingo

How to play: Each student draws a 3×3 (or 4×4) grid and fills it with any 9 (or 16) terms from the word bank below. The caller reads clues from the numbered list **in a random order** (tick them off). Players cross off the matching term. First to a line / full house calls "Bingo!" — verify against the clue numbers called.

Word bank (20 terms):

Lossless	Lossy	RLE	Dictionary compression
Symmetric	Asymmetric	Hashing	Primary key
Foreign key	Composite key	Normalisation	ACID
Record locking	LAN	WAN	Packet switching
DNS	Router	Switch	PageRank

Caller's clue list:

1. Compression where the original can be recovered exactly. *(Lossless)*
2. Compression that permanently removes data, used for JPEG and MP3. *(Lossy)*
3. Replaces runs of repeated values with a value and a count. *(RLE)*
4. Stores repeated patterns once and references them; the index must be sent too. *(Dictionary compression)*
5. Encryption using the same key to encrypt and decrypt. *(Symmetric)*
6. Encryption using a public and private key pair. *(Asymmetric)*
7. One-way function producing a fixed-length value; used to store passwords. *(Hashing)*

8. Attribute(s) that uniquely identify each record. *(Primary key)*
9. The primary key of another table, used to link tables. *(Foreign key)*
10. A primary key made from two or more attributes combined. *(Composite key)*
11. Organising data to reduce redundancy and remove anomalies. *(Normalisation)*
12. The four properties of a reliable transaction. *(ACID)*
13. Locking a record during a transaction to prevent the lost update problem. *(Record locking)*
14. A network covering a small area on a single site, hardware owned by one organisation. *(LAN)*
15. A network covering a large geographic area, using third-party infrastructure. *(WAN)*
16. Data split into independently routed units, reassembled at the destination. *(Packet switching)*
17. Translates domain names into IP addresses. *(DNS)*
18. Connects different networks together using IP addresses. *(Router)*
19. Connects devices within a LAN using MAC addresses. *(Switch)*
20. Ranks web pages by importance using incoming links. *(PageRank)*

Quick-fire 'Guess the term'

How to play: Read a clue aloud; students write or shout the answer. 12 clues, one point each — great as a starter or plenary. Keep a running total over the term.

1. The lossless compression best suited to images with long runs of identical pixels. → **RLE (Run Length Encoding)**
2. The encryption weakness that asymmetric encryption solves by sharing a key openly. → **Key distribution problem**
3. The property of a hash function meaning the same input always gives the same output. → **Deterministic**
4. The normal form that removes transitive dependencies. → **3NF (Third Normal Form)**
5. The table used to resolve a many-to-many relationship. → **Linking / junction table**
6. The ACID property meaning a transaction is all-or-nothing. → **Atomicity**
7. The rule that a foreign key must reference a valid existing primary key. → **Referential integrity**
8. The SQL keyword that filters which rows are returned. → **WHERE**
9. The SQL clause that combines columns from related tables on a matching key. → **JOIN (… ON)**
10. The TCP/IP layer that adds source and destination IP addresses. → **Network / Internet layer**
11. The protocol used only for *sending* email. → **SMTP**
12. JavaScript form validation runs in this location, which is why it is not secure on its own. → **Client-side (the browser)**